Measure sound speed

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Analyze sound travel measurements and their limitations.

SUCCESS CRITERIA - I can...

1. Define one-way versus round-trip paths.
2. Calculate speed from distance and time.
3. Use repeated measurements and discuss timing uncertainty.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Two wave displacements at one point are **-3 mm** and **+7 mm**. **Find the resultant** and state whether they reinforce or partly cancel.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: v = d/Δt; v = **2** d/Δt for an echo.
Match the distance to one-way or round-trip travel. Use moderate sound levels and **record actual data separately**.

Worked reasoning: A sound echo returns **0.4 s** after a pulse. At **340 m/s**, round-trip distance is **136 m**, so the reflector is **68 m** away.

Your turn: Draw outgoing and returning paths and **mark where the factor of 2** enters.



Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

A pulse travels between two sensors separated by **1.70 m**. Delay is **0.0050 s**. Is this one-way or echo distance? **Find speed**.



Marzano check - circle: ① ② ③

04 Apply the physics

A pulse in a wave simulation travels **12 m** one way in **5 s**.
1. **Label distance** and time, then **calculate its speed**.
2. If the distance stays fixed but the recorded time is larger, does the calculated speed rise or fall?



Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

Plan a table for three trials. Include distance, elapsed time, method (one-way or echo), speed and one possible systematic error.

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

Two equal-density spheres have radii r and **4** r. Supplied: area scales by **16**, mass by **64**. **Compare their area per mass**.
Label energy entering, leaving and stored.

Draw and label all energy transfers across the boundary.

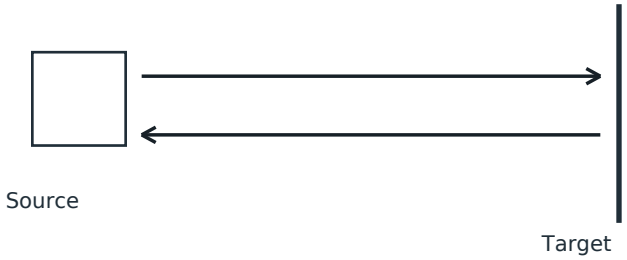
Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

07 Exit check

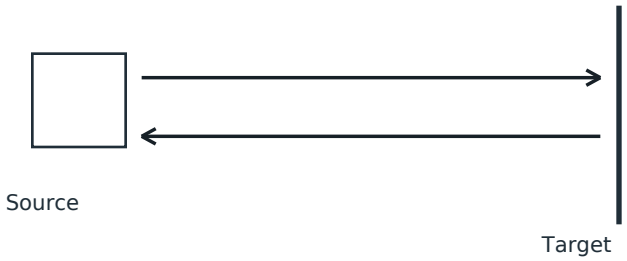
Why does using d/t instead of **2** d/t for an echo give an incorrect speed?



Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

Return to the worked example in problem **2**. Two reflectors return echoes at **0.2** and **0.6 s**. **Find both ranges** and their separation along the same line.



Marzano check - circle: ① ② ③